

UI_BOARD

Version: V0.0

28/06/2026

DRAFT

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[BLOCK_DIAGRAM.SchDoc](#)



[POWER.SchDoc](#)



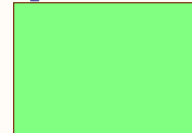
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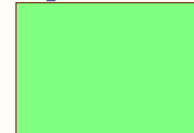
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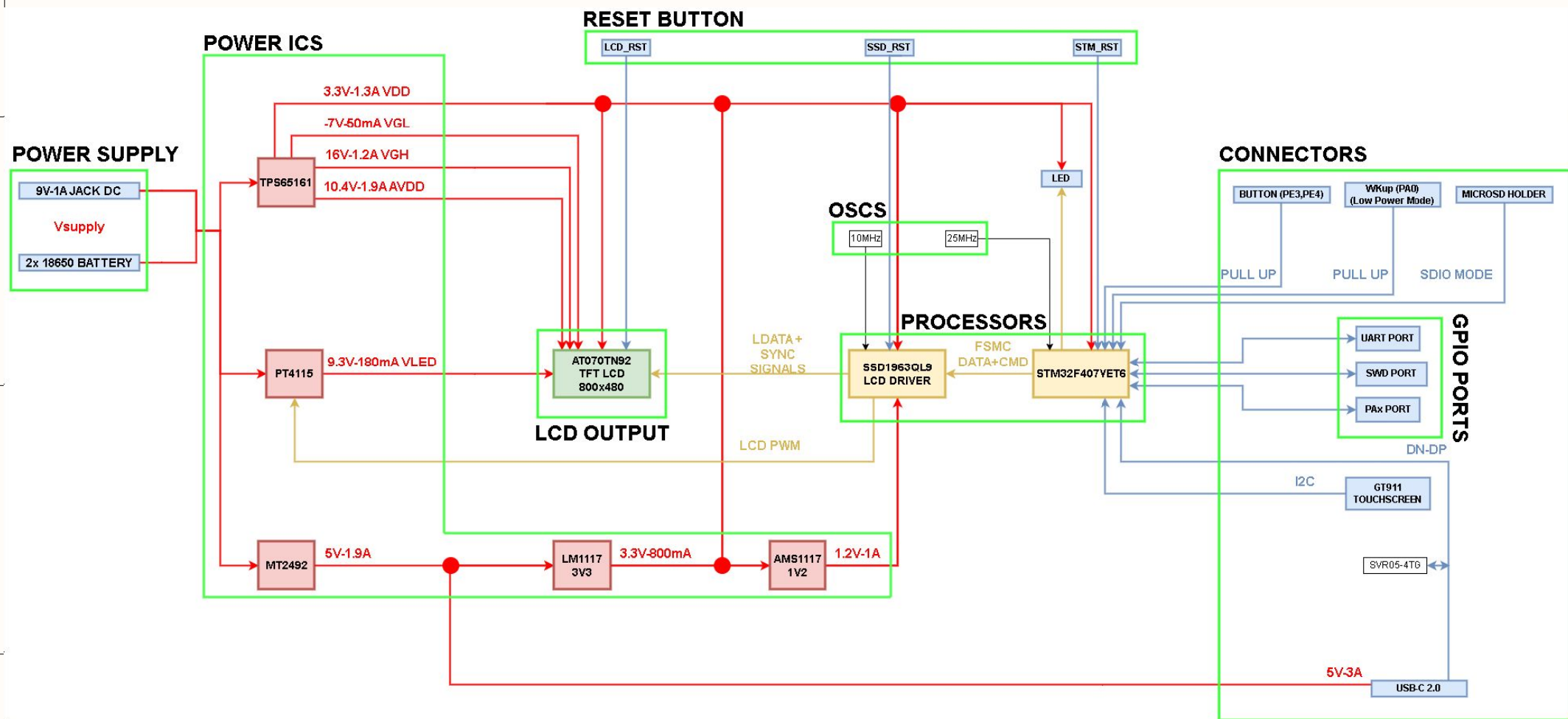


[CONNS.SchDoc](#)



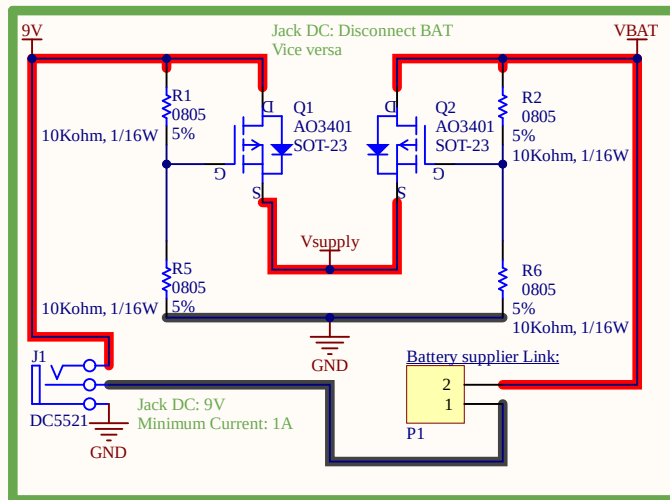
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BLOCK DIAGRAM

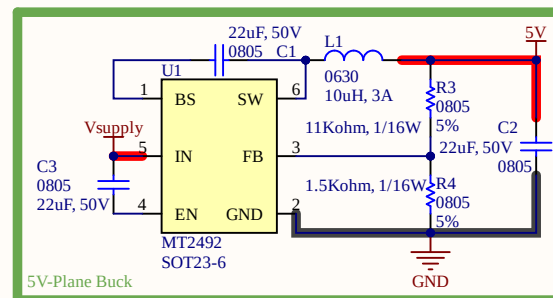


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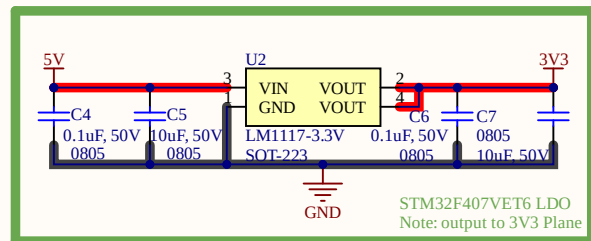
PWR_INPUT



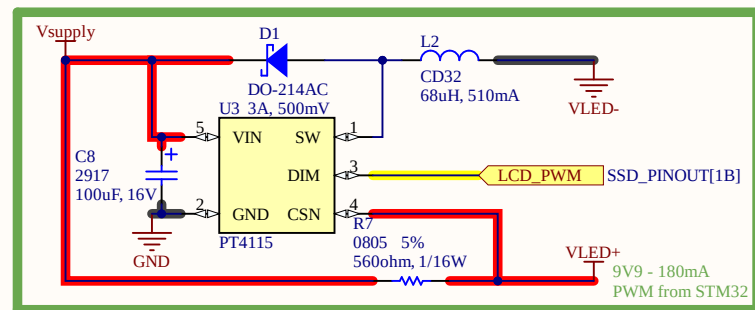
GLOBAL_5V



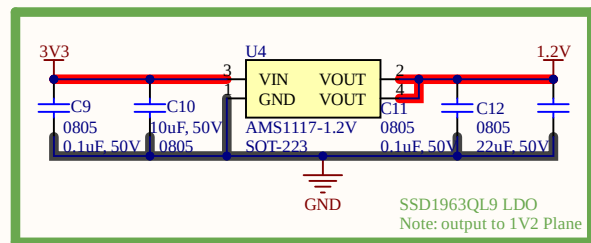
LDO_STM32



BACK_LIGHT

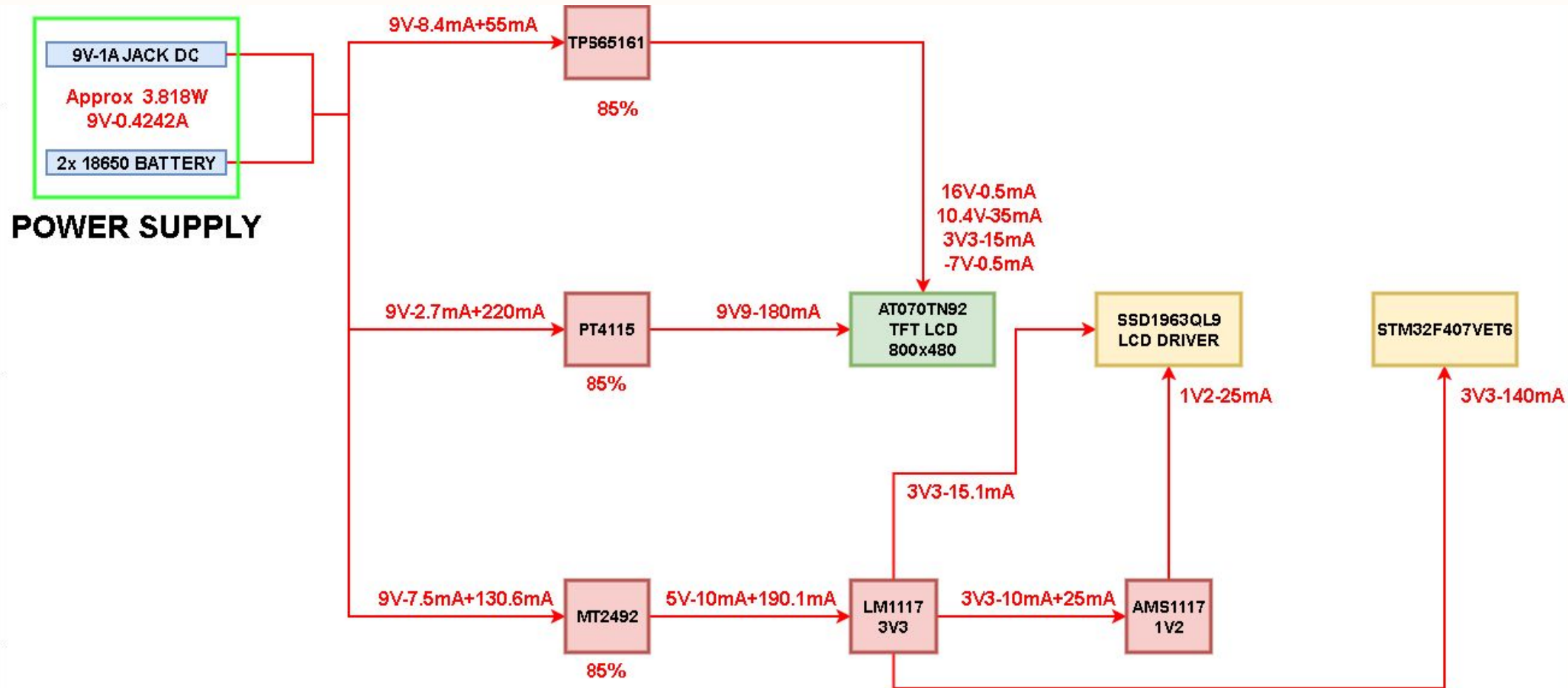


LDO_SSD1963



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POWER DIAGRAM



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The schematic diagram illustrates a power management system for a microcontroller, centered around the TPS65161 PMIC. The system is powered by a 10.4V supply and includes several key components and circuits:

- Input Supply:** 10.4V supply connected to the system.
- Negative Charge Pump Circuit:** Generates a -7V supply using a DO-214AC diode (D6) and a network of capacitors (C73, C74, C79, C80, C83, C85, C88, C89, C90, C91, C92, C93, C94, C97, C98, C99) and resistors (R36, R38, R39, R40, R41).
- TPS65161 PMIC:** The central microcontroller, configured with various capacitors (C73-C99) and resistors (R32-R41) to regulate different voltage rails. The PMIC's internal regulators are shown, with their output voltages calculated using the formula $V_o = V_{ref} \left(1 + \frac{R_{top}}{R_{bot}}\right)$.
- Boost Circuit:** A boost converter (AO3401 SOT-23) that steps up the input voltage to 10.4V.
- Buck Circuit:** A buck converter (DO-214AC) that steps down the input voltage to 3V3.
- Timing Diagram:** A graph showing the relationship between VGH, B/L, AVDD, DVDD, and DATA signals over time, with a 20ms scale.
- LED Driver:** A small circuit (D10) that drives a 3V3 LED.

The diagram is labeled "Software run LCD EX:".

```
HAL_GPIO_WritePin(GPIOE, GPIO_PIN_2, GPIO_PIN_SET);
HAL_Delay(30);
SSD1963_Init();
SSD1963_Fill_Screen(BLACK);
HAL_Delay(50);
HAL_GPIO_WritePin(GPIOB, LCD_BL_PIN, GPIO_PIN_SET);
SSD1963_Set_PWM_Brightness(255);
```

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SWD

I2C/SPI

COUPLING CAP

LED

RESET

LOW PWR WKUP

PINOUT/UART

SWD/UART

OSC|25MHz

EXT_VREF

BTN INPUT

SWD w/ SWO for FB

Note: Used by GT911 already

SPI2/I2S for SPKer

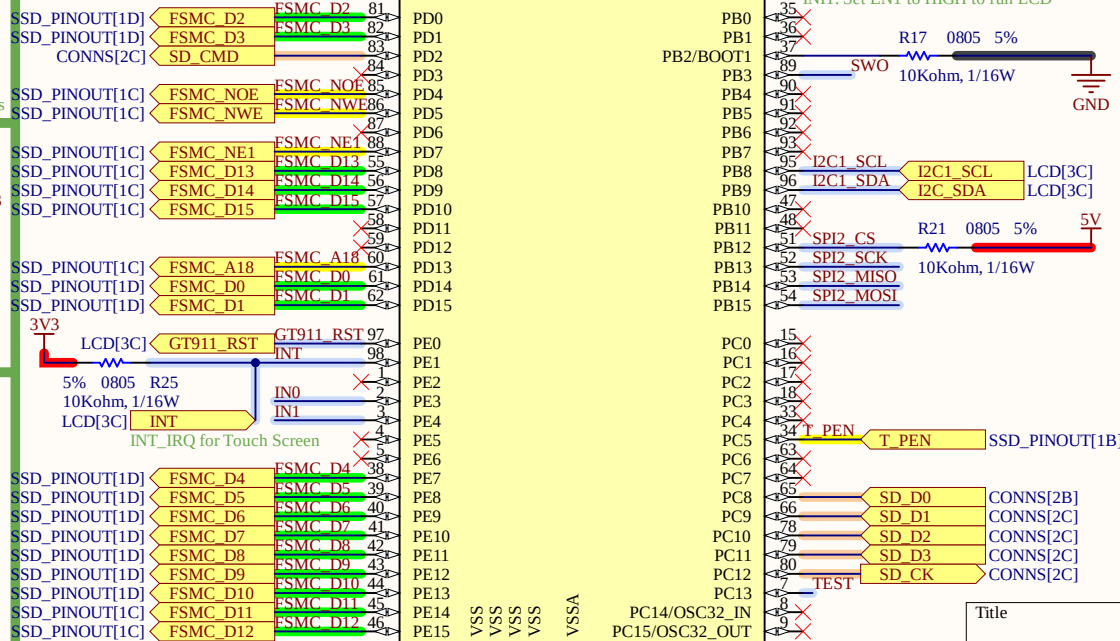
Solder ONLY 1
0: SWD
1: UART

Ex: pllm:25, plln: 36, pllp:2

Note: 3 separate Reset buttons

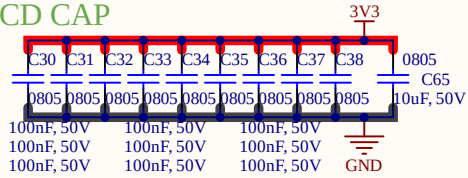
Pressed: High
Released: HiZ

Pull_UP only

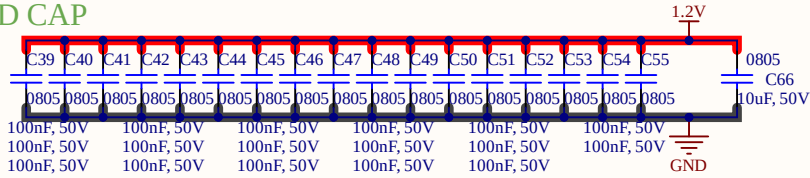


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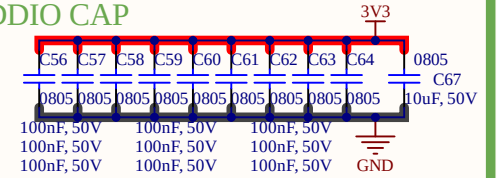
VDDLCD CAP



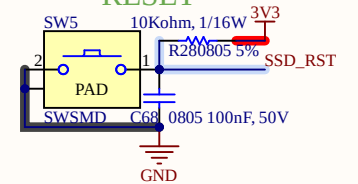
VDDD CAP



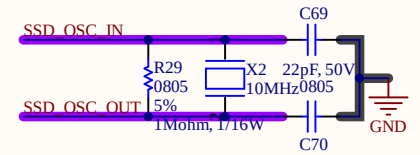
VDDIO CAP



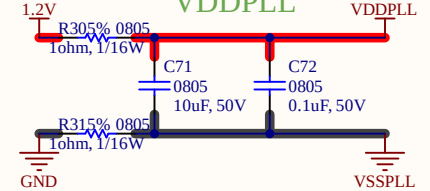
RESET



OSC|10MHz



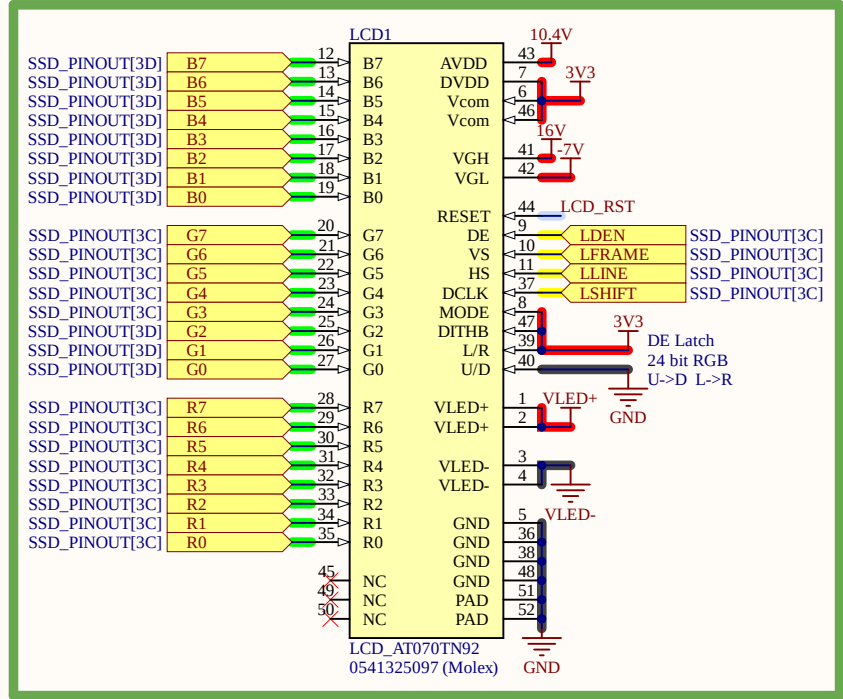
VDDPLL



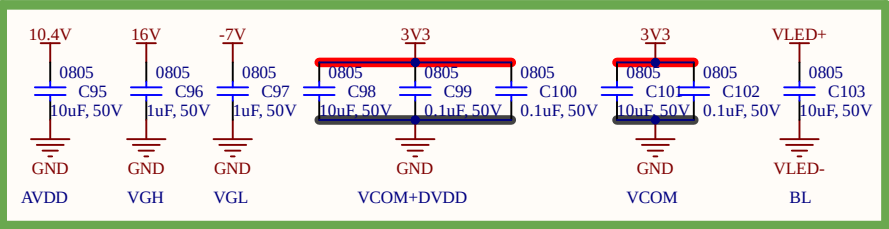
Interface	D[15]	D[14]	D[13]	D[12]	D[11]	D[10]	D[9]	D[8]	D[7]	D[6]	D[5]	D[4]	D[3]	D[2]	D[1]	D[0]
16 bits (565 format)	R5	R4	R3	R2	R1	G5	G4	G3	G2	G1	G0	B5	B4	B3	B2	B1
16 bits	R7	R6	R5	R4	R3	R2	R1	R0	G7	G6	G5	G4	G3	G2	G1	G0
	B7	B6	B5	B4	B3	B2	B1	B0	R7	R6	R5	R4	R3	R2	R1	R0
	G7	G6	G5	G4	G3	G2	G1	G0	B7	B6	B5	B4	B3	B2	B1	B0

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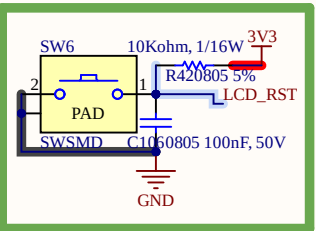
TFT_LCD



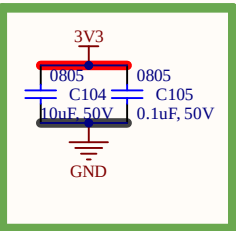
TFT_LCD_CAP



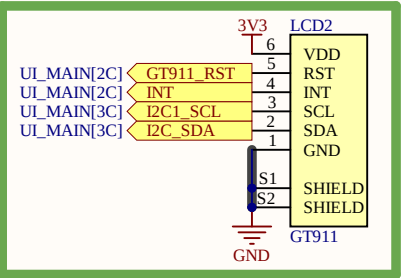
RESET



TCH_SRC_CAP

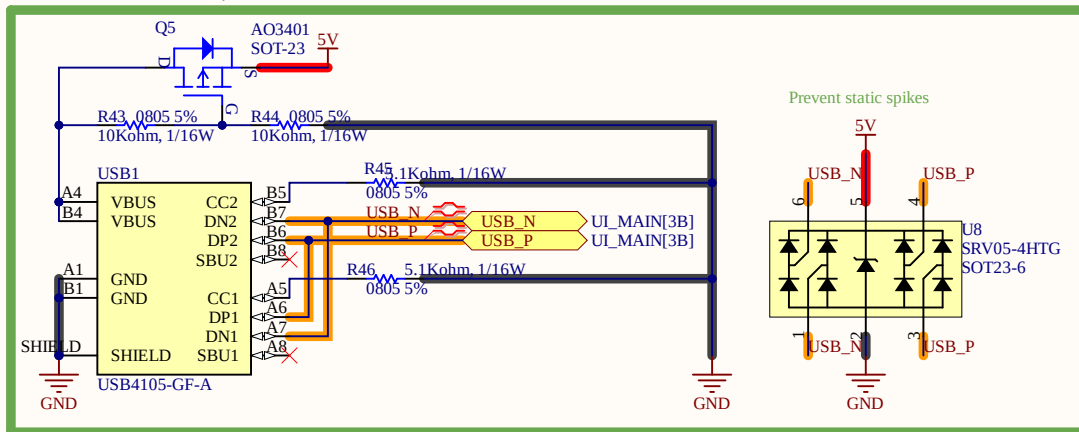


CAPACITIVE_TOUCH_SCREEN

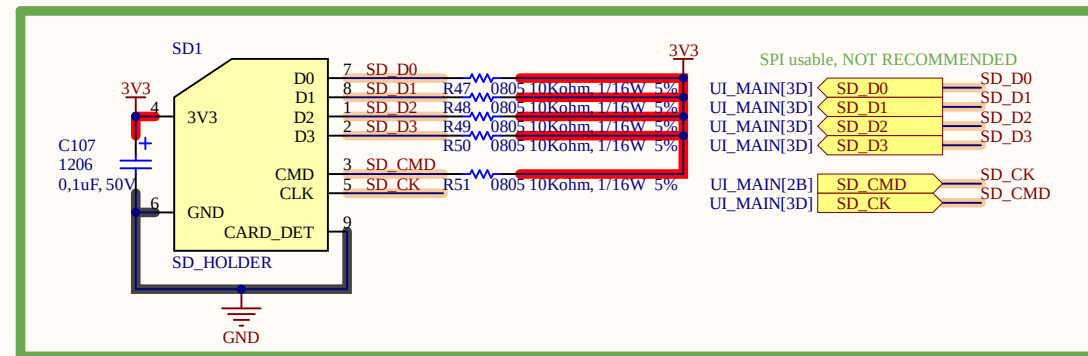


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USB-C_DN|DP



SDIO_SD-HOLDER



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